EXOTIC inits.json Default Options

Effective values used when individual initialization keys are omitted.
Current EXOTIC checkout - 9 August 2026

Important: an omitted key and a key explicitly set to `null` are not always equivalent. An omitted key preserves the loader default; an explicit `null` replaces it and may trigger inference, a prompt, or validation.

Scope and structure

The top-level `user_info`, `planetary_parameters`, and `optional_info` objects must exist, even if some individual keys inside them are omitted. The `inits_guide` object is documentation only and may be removed.

Template difference: the repository sample currently contains `multiprocess_bad_pixel_precheck = "y"`. If that key is absent, the actual loader default is disabled.

17	20	12	26
disabled settings	enabled settings	numeric defaults	unset or inferred

1. Disabled by default

These switches are effectively false when their individual keys are absent.

Initialization key	Effective default
`Fast Aperture Mask (y/n)`	false
`Ignore WCS in Header and Do Manual Alignment? (y/n)`	false
`prefer_pixel_values_over_wcs_for_target`	false
`disable vertical flux normalization`	false
`stellar_variability_only`	false
`use_exactly_the_comps_provided`	false
`use_nextastro_vsx_cache_first`	false
`detect_bad_pixels_before_photometry`	false
`multiprocess_bad_pixel_precheck`	false
`use_legacy_psf_flux`	false
`use_adaptive_apertures`	false
`use_aperture_corrections_and_full_image_fwhm`	false
`skip_low_comparison_coverage_rejection`	false
`fit_lightcurve_to_every_comparison_candidate`	false
`automatic_optimal_calibration_selector`	false
`use_ensemble_photometry_rather_than_single_comp`	false
`Use target-driven comp selection rather than comp-driven comp selection`	false

2. Enabled by default

These switches are effectively true when their individual keys are absent.

Initialization key	Effective default
`require_comp_star`	true
`use_ensemble_photometry_for_stellar_variability`	true
`require_apparent_magnitudes`	true
`photometer_fortuitous_variables`	true
`use_single_comparison_for_fortuitous_variables`	true
`detrend_on_outoftransit_baseline`	true
`use_eebls_to_initialize_tmid_and_bounds`	true
`pick_comparison_by_eebls_snr`	true
`use_deviation_from_expected_transit_in_qc`	true
`run_final_fit_phase_residual_clip`	true
`exit_at_first_qc_pass_solution`	true
`use_impactparameter_rather_than_inclination_to_fit`	true
`use_psf_photometry`	true
`use_aperture_photometry`	true
`reject_overexposed_stars`	true
`run_fast_ultranest_before_final_run`	true
`restrict_rprs_range`	true
`use_prior_rprs_when_posterior_pinned`	true
`restrict_ars_range`	true
`use_sparse_posterior_live_point_retry`	true

3. Numeric defaults

These are the effective numeric limits, thresholds, and fitting controls.

Initialization key	Effective default
`maximum_number_of_ensemble_comparisons_for_transit`	5
`maximum_number_of_ensemble_comparisons_for_stellar_variability`	5
`final_fit_baseline_duration_multiplier`	1.0
`deviation_from_expected_transit_in_qc_sigma`	5.0
`minimum number of live points for ultranest`	200
`automatic_optimal_calibration_selector_count`	10
`rprs_search_bound_max`	0.5
`restrict_rprs_range_percentage`	10.0
`restrict_ars_range_percentage`	10.0
`bad_wcs_threshold_percent`	3.0 percent
`overexposure_threshold_fraction`	0.9
`saturation_value`	FITS TELESCOP/SATURATE when available; otherwise 65535

Saturation handling: 65535 is the fallback sentinel. When the FITS header provides a recognized telescope-specific saturation or a valid SATURATE value, that header-derived value takes precedence.

4. Unset or automatically inferred

These settings do not have a single fixed initialization value. EXOTIC may infer them from FITS or pre-reduced metadata, generate them deterministically, request them, or leave the related optional calculation disabled.

Initialization key	When the key is absent
`Pre-reduced File:`	Unset
`Pre-reduced File Time Format (BJD_TDB, JD_UTC, MJD_UTC)`	Metadata or user input
`Pre-reduced File Units of Flux (flux, magnitude, millimagnitude)`	Metadata or user input
`Comparison Star used in Photometry (blank if none)`	Unset
`Filter Minimum Wavelength (nm)`	Unset
`Filter Maximum Wavelength (nm)`	Unset
`Calculate Limb Darkening Coefficients with Uncertainties? (y/n)`	Unset; standard-filter calculation or prompt as required
`pointing_rejection_sigma`	Disabled
`psf_seed_track_directory`	Unset
`colour_term`	Unset
`colour_term_error`	Unset
`colour_term_index`	Unset
`colour_term_bv`	Unset
`colour_term_bv_error`	Unset
`colour_term_bprp`	Unset
`colour_term_bprp_error`	Unset
`colour_equation_filter`	Unset
`gain_electrons_per_adu`	FITS header; otherwise 1.0
`read_noise_electrons`	FITS header; otherwise noise term omitted
`dark_current_electrons_per_second_per_pixel`	FITS header; otherwise noise term omitted
`flat_field_fractional_error`	FITS header; otherwise noise term omitted
`telescope_aperture_m`	FITS header; otherwise scintillation term omitted
`scintillation_coefficient`	FITS header; otherwise 0.09 when scintillation can be calculated
`Image Scale / Pixel Scale`	Unset
`Exposure Time (s)`	FITS/data metadata or user input
`Random Seed`	Deterministically generated from planet name and mid-transit time

5. Defaults outside optional_info

Initialization key	Effective behavior
`user_info / Add Comparison Stars from AAVSO? (y/n)`	false
`user_info / Observatory Full Title`	Empty string
`user_info / Directory of Flats`	None unless supplied
`user_info / Directory of Darks`	None unless supplied
`user_info / Directory of Biases`	None unless supplied
`user_info / Demosaic Format`	None unless supplied
`user_info / Demosaic Output`	None unless supplied
`Other user_info fields`	No fixed silent default; header lookup, inference, prompt, or validation
`planetary_parameters fields`	Initially unset; later reconciled with archive/input data
`Missing Gaia distance or proper motion`	May be populated from target coordinates

Practical rule: omit an optional key when the built-in behavior is wanted. Use an explicit value when reproducibility matters. Avoid using `null` as a generic stand-in for 'use the default' unless that specific option documents it.

Source locations

Loader defaults: `exotic/inputs.py`, `Inputs.__init__`. Recognized initialization keys: `exotic/inputs.py`, `Inputs.comp_params`. Example authoring surface: `inits.json`.

This reference describes the current local checkout and should be regenerated if the configuration surface changes.